Department of Computer Science and Engineering
Shiv Nadar University Chennai
July-November 2026 Semester
M.Tech. Course on Deep Learning
Lab Assignment 2

Date: 11th August, 2026

1. **For 2-dimensional data of two nonlinearly separable classes in Dataset1, build a multilayer feedforward neural network with two hidden layers, for each of the following configurations:**

   Configuration 1: 6 nodes in the first hidden layer and 4 nodes in the second hidden layer

   Configuration 2: 12 nodes in the first hidden layer and 8 nodes in the second hidden layer

   The model is to be trained using the pattern mode of learning, and using the generalized delta rule with $\eta = 0.8$ and $\alpha = 0.9$. Use the hyperbolic tangent function with $\beta = 1.0$ as the activation function in the hidden layers, and the logistic sigmoid with $\beta = 1.0$ as the activation function in the output layer. The stopping criterion is that the change in the average training error from one epoch to the next epoch is below 0.01.

   For each of the configurations, give the Error vs Epoch plots for (i) the training data error and (ii) the validation data error.

   For each of the configurations, plot the decision region plot or the decision boundary plot, with the training examples superimposed on it.

   For each of the configurations, determine the classification accuracy on training data and validation data, after the training is completed.

   Use the validation method to choose the best configuration. For the best configuration, determine the classification accuracy on the test data.